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Infrared Thermography and its Applications: A Review

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Abstract*— Thermal imaging camera is a device which is used to detect infrared radiations emitted from an object to give a thermal profile of the scene. This device was initially used for surveillance purpose and as night vision camera. With the advancement in technology, there are significant additions to camera capabilities and prices has decreased. Therefore, infrared thermography (IRT) has grown to become very popular and widely accepted tool as it enables temperature measurement in real time. IRT is very convenient, fast, reliable, non-contact and cost-effective method which can be practiced for condition monitoring as well as for preventive and predictive maintenance in different areas such as electrical stations, buildings surveys, mechanical components and equipments. This survey presents a review of the advancement in IRT cameras and its applications in various fields. The basics of IRT and thermal radiation are discussed in details.

Keywords- thermal camera, IRT, condition monitoring

I. Introduction

Over the last several of decades, there is a huge development in automatic vision systems. Visible light (0.4-0.7 μm) can be captured using a standard digital camera in the form of RGB or grayscale images. A digital camera captures the visible light reflected from the object surface, therefore colors and visibility of image depend upon illumination, changing color intensity etc. In absence of visible light, these cameras cannot capture pictures. To overcome some of these shortcomings, different sensors like 3D sensors and near-infrared sensors were introduced. Night vision cameras make use of the near-infrared spectrum (0.7-1.4 μm), therefore they are less dependent upon illuminance. Radiations are transmitted by the objects themselves in mid and long-wavelength infrared spectrum (3-14 μm). In this region, wavelength and intensity depend upon temperature and not upon external energy source [1]. Thermal cameras use this property and calculate the radiation in these parts of spectrum to give the thermal representation of the scene.

IRT is an approach which involves thermal image acquiring and thermal data processing from a non-contact measurement tool i.e. thermal camera. It is based on infrared radiation which is a portion of the electromagnetic spectrum with wavelengths longer than visible light. Whenever the temperature of an object is greater than absolute zero (i.e. $T > 0K$), it emits infrared radiation. The human eye can only see visible light. Thus, thermal cameras are required which acquires infrared radiation emitted by an object, converts it into temperature and display the thermal image. Measured temperature value depends upon amount of radiations emitted from an object. Higher the intensity of radiations, greater will be the temperature of an object. The advantage of a thermal camera is that it can take pictures in complete darkness [2]. Fig. 1 (a) and (b) shows example of digital image and thermal image of same printed circuit board (PCB).

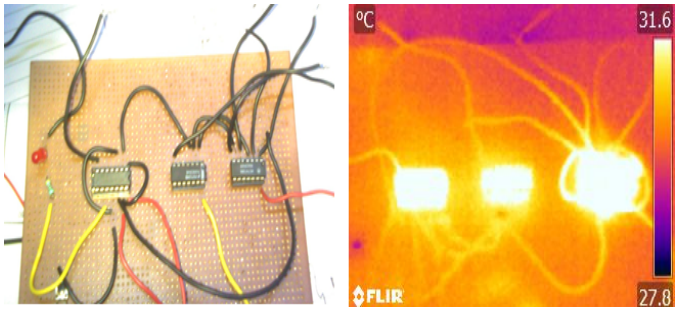

Fig. 1. (a) Digital Image (b) Thermal Image

Detector is main part of the thermal camera. In last few years, with commercialization in detector technology, thermal cameras are now available with more features and at lower prices. Therefore, IRT is now being adopted in many applications by electronic industries, medical science, surveillance and agriculture [8-24]. The remaining paper is framed as follows: Section 2 gives basic understanding about thermal radiations. Section 3 explains about fundamental components of thermal camera and its types. Section 4 describes features, advantages and limitations of IRT. Section 5 tells various applications of thermal camera in different fields and finally section 6 is conclusion.

II. Basics of Thermal Radiations

The electromagnetic spectrum is consisting of several bands according to the wavelength. In 1800, Sir William Herschel discovered the existence of infrared radiations. Infrared region is a part of electromagnetic spectrum lies between visible light and microwaves (0.7-1000 μm). This infrared spectrum can be

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further divided into sub-regions as shown in table 1. Most of radiations are blocked by atmosphere. In MWIR and LWIR, intensity of emissions and atmospheric transmittance is higher than other bands therefore these two bands are most useful for IRT applications. Atmospheric transmittance for different wavelengths bands is illustrated in Fig.2. Between 5-7.5 μm, due to atmospheric absorption, no radiation is transmitted. Maximum transmittance is observed in region of MWIR and LWIR [2].

TABLE I
DIFFERENT SUB-REGIONS OF INFRARED SPECTRUM

Sub-Region Name	Abbreviation	Wavelength (um)
Near-infrared	NIR	0.7-1.4
Short-wavelength infrared	SWIR	1.4-3.0
Mid-wavelength infrared	MWIR	3.0-8.0
Long-wavelength infrared	LWIR	8.0-15.0
Far-infrared	FIR	15.0-1000

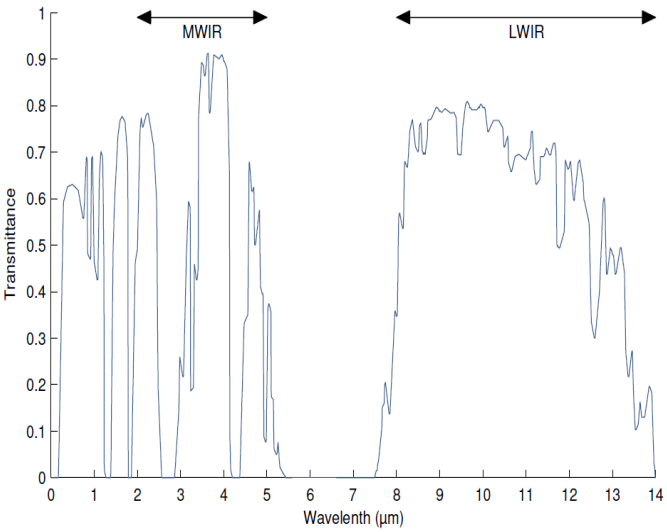

Fig. 2. Atmospheric transmittance in sub-regions of infrared spectrum [2]

The radiations emitted by blackbody at temperature T can be measured by using Plank's law i.e. power emitted per unit area per unit wavelength is given by equation 1 as,

$$W_{\lambda b} = \frac{C_1 \lambda^{-5}}{e^{\frac{C_2}{\lambda T}} - 1} \tag{1}$$

where T is the temperature (K), λ is the radiation wavelength (μm), C_1 and C_2 are constants. The wavelength of electromagnetic radiation depends on the temperature of the object; wavelength is inversely proportional to temperature. For specific temperature, wavelength of peak intensity can be calculated by differentiating equation 1 with respect to λ called as Wien's law.

$$\lambda_{peak} = \frac{0.0029}{T} \tag{2}$$

On integrating Plank's law over all the wavelengths λ , we can calculate total blackbody radiations as shown in equation 3.

$$W_{\lambda} = \sigma T^4 \tag{3}$$

σ denotes Stefan-Boltzmann constant ($5.670\times 10^{-8}) W/m^2K^4$, this is called as Stefan-Boltzmann formula. Equation 3 is applicable only for perfectly black body. In general for greybodies, equation 3 can be written as ,

$$W = \varepsilon \sigma T^4 \tag{4}$$

where ε is the emissivity of the object. For different materials values of emissivity varies between 0 and 1 [2].

III. THERMAL IMAGING CAMERA

Infrared spectrum was investigated in 1800, but first infrared imaging device was introduced in 1947 for military purposes as night vision camera. From then thermal cameras has evolved. Nowadays different application-specific thermal cameras are available in market. Thermal cameras have become compact in size, easy to use and very affordable. Different performance parameters are used to choose right camera for particular application and to exploit its ability. These parameters include Spectral range, Detectivity, Temperature range, Absolute accuracy, Frame rate, Spatial resolution and Thermal sensitivity, Environment or operating conditions [3].

A. Thermal Camera Types

There are basically two ways in which thermal camera can be used, first is scanning devices and second is 2-D IR focal plane array (IRFPA). As a scanning device, camera captures single point or row at a time. As IRFPA, it captures all elements at a same time with every element in array. Detector is heart of thermal camera. There are 2 main types of optical detectors used:

1) *Photon Detectors* : In these types, radiation is absorbed within material by interaction between electrons. Photon detectors works in MWIR band. They show good signal-to-noise performance and they are very sensitive to small variations in temperature but to achieve this they require cryogenic cooling. Liquid nitrogen and cryocooler is used for cooling, thus they are costlier than thermal detectors.

2) *Thermal Detectors* : In these types of detectors, material temperature is changed due to absorption of incident radiation. Thus, material properties changes and electrical output is generated. Thermal effects are independent of wavelength. Thermal detectors work in LWIR band. Cryogenic cooling is not required for thermal detectors. Three basic types of thermal detectors are as follows:

a) *Thermopile*: It is collection of thermocouples. Better temperature sensitivity is achieved by connecting thermocouples in series.

b) *Pyroelectric detectors*: Variation in the internal electrical polarization is calculated.

c) *Bolometers*: Variation in the electrical resistance is converted into electrical output which is processed into an image [4].

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B. Thermal Camera Lens

The digital and thermal camera is mainly differentiated by its lens. Glass allows negligible amount of infrared radiation to pass through it. Therefore, glass cannot be utilized in thermal cameras. Germanium has high transmittance for thermal radiations, thus used as lens material for thermal camera. Germanium has a high price, therefore thermal cameras are costly. f number is the ratio of the focal length of a camera lens to the diameter of the aperture being used. The price of lens depends upon f number. Greater the f number value, price of camera will be less [5].

C. Thermal Camera Output

Nowadays, thermal cameras available in market are very handy and have same size and shape as digital camera. Grayscale format is used to represent images with a depth of 8 to 16 bits per pixel [1]. To make thermal images easy to interpret and analyze, thermal images are represented using a pseudo colors to highlight the difference in temperature. IRON is the name of most common color palette used in thermal camera, in which black color is used to indicate minimum temperature, then blue and purple color for slightly higher temperature, the mid-range of temperatures is normally red, orange and yellow and white color for maximum temperature. Thermal image includes a color-temperature scale which shows the different colors used for various temperature ranges. These pseudo colors system is use just to distinguish cool regions from hotter regions. Different color pallets like LAVA, RAINBOW are shown in Fig.3.

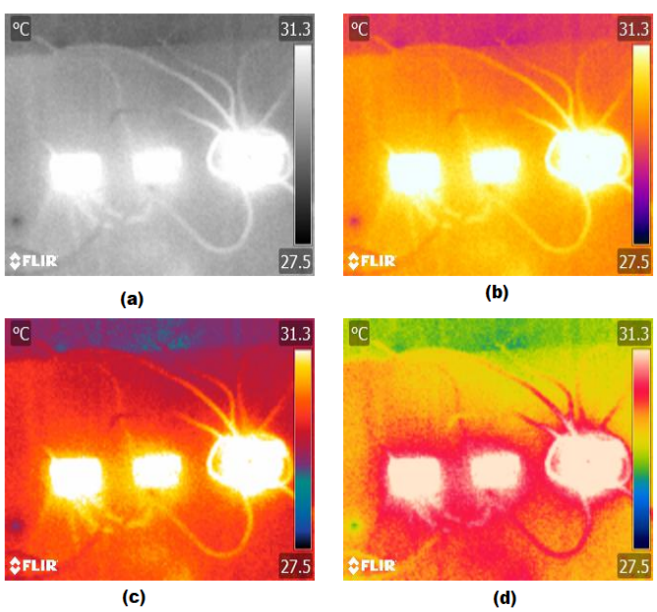

Fig. 3. Different pallets available in thermal camera (a) GRAYSCALE (b) IRON (c) LAVA (d) RAINBOW

IV. Thermal Camera Features

Today's thermal imaging devices have very advanced features, therefore IRT has become an efficient tool for diagnosing faults and preventive maintenance in various fields. Some of these advanced features are as follows:

1) Camera Resolution: The thermal imaging cameras comes with 320×240 or 640×480 pixels resolution. Due to higher resolution, we can get clear thermal images which helps in accurate analysis. Higher resolution helps to cover larger object with single image with better clarity.

2) Thermal Sensitivity: Thermal cameras can detect smallest changes in temperature because of higher thermal sensitivity. Advanced thermal camera comes with thermal sensitivity of 0.03^0 Celsius.

3) Accuracy: Accuracy indicates how precisely thermal camera measures the temperature value. The advanced thermal camera shows accuracy of ± 1 %.

4) Manual Span and Level Correction: Thermal camera allows to set temperature range manually. It helps to measure small temperature details of particular area.

5) Thermal Fusion: This feature allow to combine digital and thermal image smoothly. It helps to locate and isolate problems easily.

6) Laser Pointer and Interchangeable lens: With laser pointer user can know where camera's lens is focused, avoid direct contact with hot regions. Interchangeable lenses allows to change lens according to application. Other features include compatibility with external tools and measurement devices, quick data transmission through cable, Wi-Fi, Bluetooth, addition of speech and text comments. With inclusion these features thermal camera has become more effective tool [6].

A. IRT Advantages and Limitations

IRT has many benefits over other technologies like, it does not emit any harmful radiations, therefore it can be used multiple times for longer durations. IRT is a non-invasive method, therefore it does not encroach upon the object in any way. IT gives both thermal and digital images therefore it is easy to analyse and locate the fault locations. IRT helps to find heated regions and deteriorating equipments prior to failure, which helps in saving lots of repair time and cost [2]. However, there are some drawbacks in IRT method. Sometimes thermal images are complex to understand in specific objects having inconsistent temperatures. Due to different emissivity of different objects and reflections from shiny surfaces, sometimes temperature reading might not be precise as expected. Thermal imaging cameras are unable to measure the temperature of underwater objects. Thermal imaging cameras cannot see through a glass and wall. They can measure only surface temperature [7].

V. IRT Applications

Temperature is one of the most reliable parameter which gives thermal distribution and tells about condition of equipments using these thermal data. Basically, IRT is used for condition monitoring, inspections, fault diagnosis and preventive maintenance. This section will discuss about thermal camera applications in various fields.

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A. Condition Monitoring of Electrical Systems

Electrical systems operate at high voltages continuously. The reliability of these systems depends upon stable temperature in all equipments. Internal heat may rise due to various reasons like loose connections, corrosion, overvoltage, load imbalance, defects in parts, insulation problems etc. Faults in electrical equipments may lead to abnormal rise in temperature and results in unplanned outages. IRT can be used as diagnostic tool to measure temperature of equipments, locate the faulty parts and extent of heat spread. Testing during normal operation conditions can be dangerous and shutting down a station is very costly. With thermal camera, inspection can be done without interrupting normal working of system. IRT is non-contact, so it is very safe and efficient compare to other techniques.

If the equipment is operated at a high-power rating, it causes excess heating in equipment and reduces its efficiency. The main reason behind rise in maintenance cost is the lack of knowledge about the maintenance scheduling and kind of repair needed for the equipment. If operating performance of electrical systems inspected regularly, then early preventive maintenance can be achieved and this increases lifetime of equipment. Also, the unnecessary maintenance expenses can be saved. Fig. 4 shows thermal image of Light distribution board showing hotspot on B phase. Possible cause behind heating is loose connection.

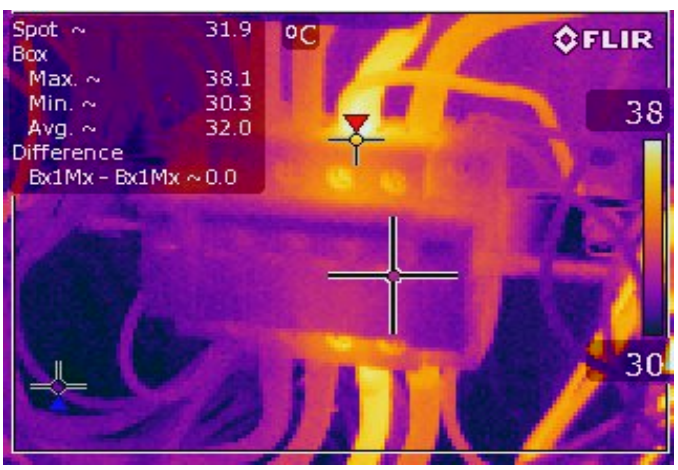

Fig. 4. Thermal Image of Light Distribution Board

IRT can be used with applications of neural network as preventive and predictive method [8][9]. By using different image processing techniques IRT has become efficient tool for automatic fault diagnosis [10-13]. Recent progress in the field of circuit technology has resulted in enormous increase in usage of PCB on which millions of small electronic circuits are present. As PCB is integral part of most of electrical circuits, fault identification and localization is important to achieve expected performances. IRT can be applied to detect abnormal heating on PCB [14-16].

B. Gas Detection

We cannot see some of chemical compounds and gases by the naked eyes. Some harmful, poisonous gases pollute air. Chemical compounds and gases from oil refineries and petrochemical plants passes through pipes, valves. IRT gives complete information about the condition of pipe, tube and valve insulation. In petrochemical facilities, there may be many leak spots. Thermal cameras can be employed to detect gas leakage. Thermal cameras can capture selective infrared bands. When radiations from these bands are absorbed by gas, the particular area would appear as dark if the gas is present [1]. The camera gives a representation about condition of these equipments and leaks can be visualised as smoke on the camera screen and user can find out presence of harmful gases. This gas detection can be achieved in real time. By using IRT, inspectors can safely detect gas leakage without getting exposed to harmful chemicals [17]. With thermal camera volcano ash particles can be detected.

C. Civil Structures Inspections

Bridges, roads, airports and buildings are main parts of our daily lives. Therefore, information about condition of these structures is very essential as sudden accidents can happen which may lead to loss of human lives. By using IRT, user can scan buildings, structures thoroughly in very short time. Thermal camera can find out areas of energy leaks near windows, doors and can also find roof leakages. IRT can be used to find out cracks, holes, moisture in walls, air leaks, badly insulated areas, faulty electrical supply lines. These problems are mostly not visible to naked eyes. Regular inspections of building and civil structures can helps to finding out problems in early stages which helps to avoid serious hazards. Therefore, IRT becomes important safety tool and can save valuable time and money [18] [24]. Fig. 5 shows thermal image of building. Red spots are present near windows and doors regions indicates heat loss while blue color shows cooler regions.

Fig. 5. Thermal Image of Residential Building [24]

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IRT can be used in energy efficient smart buildings using Internet of Things (IoT). For instance thermal cameras can capture the images of auditorium to find human crowd density to turn on the fans only in the areas having human clusters and IoT devices would control the fans accordingly thereby saving energy in smart buildings [19].

D. Animals

Warm blooded animals try to maintain constant body temperature according to surrounding weather. Blood circulation and respiration of body is the reason for uneven temperature distribution over entire body. Thermal camera can be used on animals to find out hot and cold spots on their bodies. From these animal doctors can predict the problems in joints, veins and skin of animals without performing unnecessary surgery. IRT can be utilized as a diagnostic tool for finding out some diseases of animals. When dairy cattle get ill, temperature in gluteal areas rises. Hotspots on horse's body is a sign of inflammation or increased blood flow, whereas cold spots indicate sign of swelling or scar tissues. Thermal cameras can work in complete darkness, therefore it can be utilized to tracking animals in forest during night [1].

E. Agriculture

IRT has many applications in agriculture and food industries. With thermal camera, soil water level can be predicted and irrigation can be scheduled. IRT can be helpful with pathogen and disease detection. When thermal image of leaf shows rise in temperature on surface, its sign of pathogen development. Thermal cameras can help in determining fruit yield, maturing time of fruits, detection of bruises on fruits. Temperature of bruised fruit is found to be lower than unbruised fruit. To find out foreign particle in fruits or vegetable Thermosensorik CMT 384 thermal camera can be used. Difference between emissivity and difference between heat capacities of actual fruit and foreign particle is used. This method is also useful in process of grains storage by finding wet areas [20].

F. Surveillance

IRT is very useful for surveillance and detection of intruders. There are technologies like Closed-Circuit Television (CCTV), Radio Frequency Intruder Detection (RAFID) systems which are presently being used for surveillance. IRT is one excellent tool to detect intruders because thermal camera can work in complete darkness. Based on slight changes in temperature, heat signatures are formed which helps to obtain clear thermal images. IRT can also work in presence of fog and smoke. Thermal camera can detect person in daylight as well as person hiding in shadow or bushes. For detection of intruder's shape and size, pattern detection algorithms are used with thermal images. IRT can be used in following area for surveillance: Government premises, Public facilities, studio, schools, colleges, Commercial complexes, warehouses, Factories, hotels, residential areas, airports, railway stations, nuclear stations, pipelines, bridges, Military camps [1][21].

G. Medical Applications

Abnormal body temperature is reliable indicator of illness. IRT is non-contact, non-invasive tool for measurement of body temperature remotely. When IRT is used for detection of cancer, existence of cancer cells causes abnormal heat pattern in that region. Due to presence of tumor, there is increased blood flow and metabolic rate which causes increase in temperature and confirms cancer. Changes in temperature of skin surface is a sign of vascular disorders and diabetic neuropathy. Generally, Skin diseases cause inflammation which results in unusual temperature pattern on skin surface. IRT can also be applied to detect gynecological problems, skin infections, heart, fever screening and brain imaging. Thermal camera acts as an early indicator of heart attacks and also useful in diagnosis of kidney and liver diseases. By using IRT in different applications, if the disease is detected in early stages then it can be very helpful in treatment and can save lives [22].

H. Aviation

IRT can be used for safety and security in the field of aviation. Thermal camera can be used for inspection of different parts of aircraft to find out defects with high accuracy and within short period of time. Thermal cameras can be mounted on front lower side of airplane which gives pilot valuable information about weather conditions outside. Thermal camera works in night, fog, rain so it enhances the vision system and aware pilot about situation which in turn helps in safer landing of air plane [23][24].

I. Nuclear Industries

In the field of nuclear industry, IRT has been utilized for condition monitoring of nuclear equipments. Different parts like fan bearing, motors, water cooling pump can be examined to detect the presence of faults or any kind of overheating. Heavy components like main transformer, switchyard is present in a nuclear power plant. Regular monitoring of these components is important for security purposes. Condition monitoring of heavy water leakage and operational nuclear reactor can be done by IRT. For ammonia cracker maximum operating temperature is 993K. If the temperature passes this limit it may cause loss of flexibility and creep damage it may result rupture of tubes. With thermal camera surface temperature of cracker can be measured and rise in temperature can be find out. Monitoring of several safety-critical systems of nuclear power stations using IRT will assure reliability of the systems and early prevention can avoid serious damage [24].

J. Other Applications

IRT can be use in paper industry. IRT also has its applications in marine industry. Thermal camera is used for monitoring of electrical and mechanical systems on ships, mobile marine systems which helps in monitoring the presence of water pollution or leakage of poisonous materials over water surface [24]. IRT is very important tool in fire-fighting applications User can identify if a spot fire is actually out, or just about to reignite.

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VI. CONCLUSION

Over the years IRT has emerged as an effective monitoring and fault diagnostic tool for real-time temperature monitoring of targets in a non-contact, non-invasive way. In this paper, IRT and its different applications are reviewed. It presents relevant information concerning the health and effectiveness of equipment or processes which helps in early fault detection in turn, avoids hazardous accident. This article will be useful for understanding concepts of IRT, camera utility and the applications in various fields. Due to its enormous advantages, IRT can be develop further and will become more robust tool in future.

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